Ex.No:08

Capture and Analyze the packet traces of DHCP and ICMP using Wireshark

Aim

To capture and analyzes the packet traces of DHCP and ICMP protocol using Wireshark

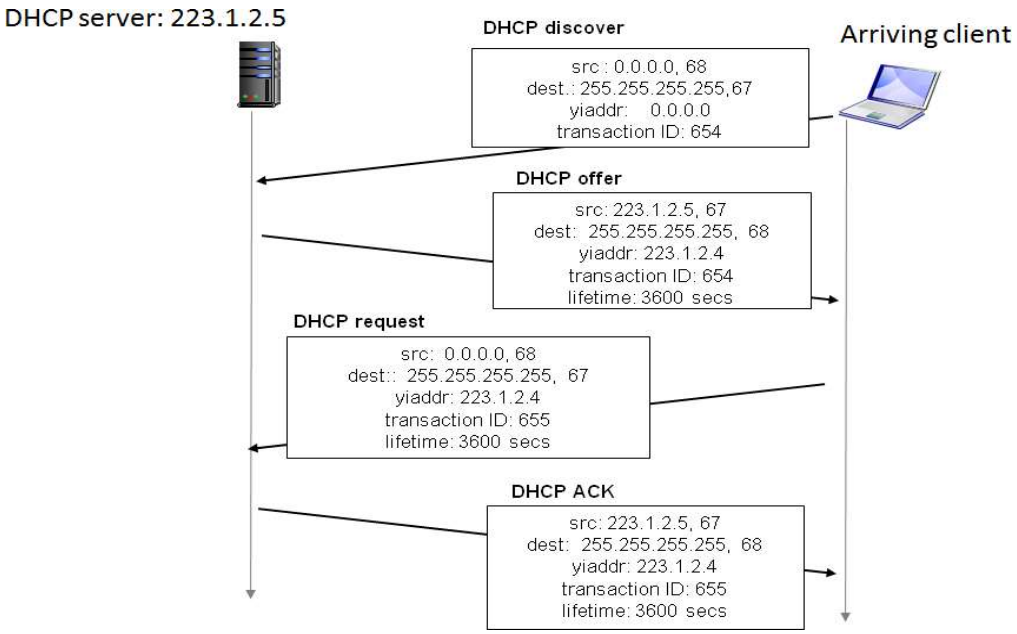
DHCP Protocol

- DHCP – Dynamic Host Configuration Protocol
- DHCP is the protocol used to assign IP address for a newly arrived client system in a network
- In addition to IP address assignment, DHCP also allows a host to learn additional information, such as its subnet mask, the address of its first-hop router (often called the default gateway), and the address of its local DNS server
- DHCP protocol is called as plug-and-play protocol or zeroconf (zero-configuration) protocol

DHCP Messages

- Assignment of IP address to a newly arrived client is done using following 4 steps:

1. DHCP Discover
2. DHCP Offer
3. DHCP Request
4. DHCP ACK

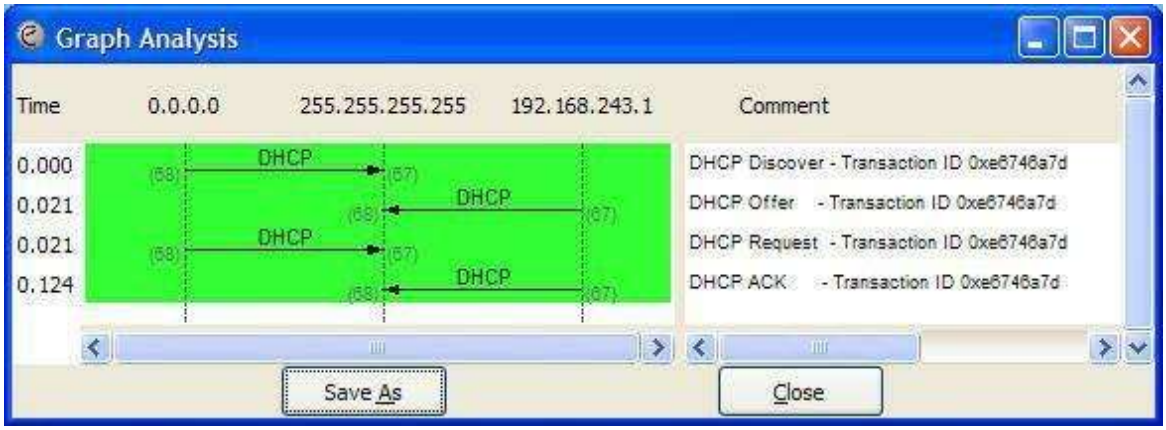


1. Execute the following commands in the command prompt to release and renew the IP address assigned to the system. During IP address renewal, the new IP address is assigned with the help of DHCP protocol

ipconfig /release ipconfig /renew

2. DHCP messages are sent over UDP (User Datagram Protocol)

```
Frame 2 (342 bytes on wire, 342 bytes captured)
Ethernet II, Src: 192.168.243.92 (00:90:4b:69:dd:34), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
    Source port: bootpc (68)
    Destination port: bootps (67)
    Length: 308
Checksum: 0xd1a1 [correct]
Bootstrap Protocol
```



3. The Link Layer address of my workstation is: 00:90:4b:69:dd:34

```
Frame 1 (342 bytes on wire, 342 bytes captured)
Ethernet II, Src: 192.168.243.92 (00:90:4b:69:dd:34), Dst: Broadcast
(ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Source: 192.168.243.92 (00:90:4b:69:dd:34)
    Type: IP (0x0800)
Internet Protocol, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
Bootstrap Protocol
```

4. The values which differentiate the Discover message from the Request message are in "Option 53: DHCP Message Type"

Frame 1 (342 bytes on wire, 342 bytes captured)
Ethernet II, Src: 192.168.243.92 (00:90:4b:69:dd:34), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Internet Protocol, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67) Bootstrap Protocol
Message type: Boot Request (1) Hardware type: Ethernet Hardware address length: 6 Hops: 0
Transaction ID: 0xe6746a7d Seconds elapsed: 1280
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0 (0.0.0.0)
Your (client) IP address: 0.0.0.0 (0.0.0.0)
Next server IP address: 0.0.0.0 (0.0.0.0)
Relay agent IP address: 0.0.0.0 (0.0.0.0)
Client MAC address: 192.168.243.92 (00:90:4b:69:dd:34) Server host name not given
Boot file name not given Magic cookie: (OK)
Option 53: DHCP Message Type = DHCP Discover Option 116: DHCP Auto-Configuration (1 bytes) Option 61: Client identifier
Option 50: Requested IP Address = 192.168.243.92 Option 12: Host Name = "homelt"
Option 60: Vendor class identifier = "MSFT 5.0" Option 55: Parameter Request List
End Option Padding

Frame 3 (350 bytes on wire, 350 bytes captured)
Ethernet II, Src: 192.168.243.92 (00:90:4b:69:dd:34), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Internet Protocol, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67) Bootstrap Protocol
Message type: Boot Request (1) Hardware type: Ethernet Hardware address length: 6 Hops: 0
Transaction ID: 0xe6746a7d Seconds elapsed: 1280
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0 (0.0.0.0)
Your (client) IP address: 0.0.0.0 (0.0.0.0)
Next server IP address: 0.0.0.0 (0.0.0.0)
Relay agent IP address: 0.0.0.0 (0.0.0.0)
Client MAC address: 192.168.243.92 (00:90:4b:69:dd:34) Server host name not given
Boot file name not given Magic cookie: (OK)
Option 53: DHCP Message Type = DHCP Request Option 61: Client identifier
Option 50: Requested IP Address = 192.168.243.92
Option 54: Server Identifier = 192.168.243.1 Option 12: Host Name = "homelt"
Option 81: FQDN
Option 60: Vendor class identifier = "MSFT 5.0" Option 55: Parameter Request List
End Option

5. The DHCP client and server both use 255.255.255.255 as the destination address. The client uses source IP address 0.0.0.0, while the server uses its actual IP address as the source. The IP address of the DHCP server is 192.168.243.1

SNo.	Time	Source	Destination	Protocol	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	DHCP Discover - Transaction ID 0xe6746a7d
2	0.020995	192.168.243.1	255.255.255.255	DHCP	DHCP Offer - Transaction ID 0xe6746a7d
3	0.021346	0.0.0.0	255.255.255.255	DHCP	DHCP Request - Transaction ID 0xe6746a7d
4	0.124018	192.168.243.1	255.255.255.255	DHCP	DHCP ACK - Transaction ID 0xe6746a7d

6. The DHCP server offered the IP address 192.168.243.92 to my client machine. The DHCP message with "DHCP Message Type = DHCP Offer" contained the offered IP.

```
Frame 2 (590 bytes on wire, 590 bytes captured)
Ethernet II, Src: 192.168.243.1 (00:08:da:50:49:c5), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol, Src: 192.168.243.1 (192.168.243.1), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
Bootstrap Protocol
    Message type: Boot Reply (2)
    Hardware type: Ethernet
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xe6746a7d
    Seconds elapsed: 1280
    Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0 (0.0.0.0)
    Your (client) IP address: 192.168.243.92 (192.168.243.92)
    Next server IP address: 0.0.0.0 (0.0.0.0)
    Relay agent IP address: 0.0.0.0 (0.0.0.0)
    Client MAC address: 192.168.243.92 (00:90:4b:69:dd:34)
    Server host name not given
    Boot file name not given
    Magic cookie: (OK)
    Option 53: DHCP Message Type = DHCP Offer
    Option 1: Subnet Mask = 255.255.255.0
    Option 54: Server Identifier = 192.168.243.1
    Option 51: IP Address Lease Time = 3 days
    Option 6: Domain Name Server = 192.168.243.1
    Option 5: Name Server = 24.29.103.10
    Option 15: Domain Name = "nyc.rr.com"
    Option 31: Perform Router Discover = Enabled
    End Option
    Padding
```

7. In my experiment, the host requests the offered IP address in the DHCP Request message.

```
Frame 3 (350 bytes on wire, 350 bytes captured)
Ethernet II, Src: 192.168.243.92 (00:90:4b:69:dd:34), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
Bootstrap Protocol
    Message type: Boot Request (1)
    Hardware type: Ethernet
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xe6746a7d
    Seconds elapsed: 1280
    Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0 (0.0.0.0)
    Your (client) IP address: 0.0.0.0 (0.0.0.0)
    Next server IP address: 0.0.0.0 (0.0.0.0)
    Relay agent IP address: 0.0.0.0 (0.0.0.0)
    Client MAC address: 192.168.243.92 (00:90:4b:69:dd:34)
    Server host name not given
    Boot file name not given
    Magic cookie: (OK)
    Option 53: DHCP Message Type = DHCP Request
    Option 61: Client identifier
    Option 50: Requested IP Address = 192.168.243.92
    Option 54: Server Identifier = 192.168.243.1
    Option 12: Host Name = "homelt"
    Option 81: FQDN
    Option 60: Vendor class identifier = "MSFT 5.0"
    Option 55: Parameter Request List
    End Option
```

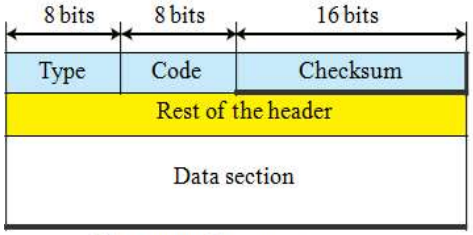
8. Finally Server sends the DHCP ACK for the client request. DHCP ACK message also contains other information like Relay agent IP address, Subnet Mask, Default Gateway(Option 3), IP Address Lease time etc

```
Frame 4 (590 bytes on wire, 590 bytes captured)
Ethernet II, Src: 192.168.243.1 (00:08:da:50:49:c5), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol, Src: 192.168.243.1 (192.168.243.1), Dst: 255.255.255.255 (255.255.255.255)
User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
Bootstrap Protocol
    Message type: Boot Reply (2)
    Hardware type: Ethernet
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xe6746a7d
    Seconds elapsed: 1280
    Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0 (0.0.0.0)
    Your (client) IP address: 192.168.243.92 (192.168.243.92)
    Next server IP address: 0.0.0.0 (0.0.0.0)
    Relay agent IP address: 0.0.0.0 (0.0.0.0)
    Client MAC address: 192.168.243.92 (00:90:4b:69:dd:34)
    Server host name not given
    Boot file name not given
    Magic cookie: (OK)
    Option 53: DHCP Message Type = DHCP ACK
    Option 54: Server Identifier = 192.168.243.1
    Option 51: IP Address Lease Time = 3 days
    Option 1: Subnet Mask = 255.255.255.0
    Option 3: Router = 192.168.243.1
    Option 6: Domain Name Server = 192.168.243.1
    Option 5: Name Server = 24.29.103.10
    Option 15: Domain Name = "nyc.rr.com"
    Option 31: Perform Router Discover = Enabled
    End Option
```

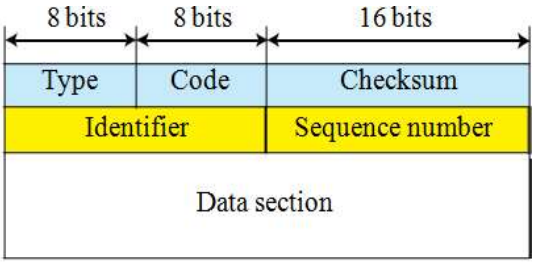
ICMP Protocol

- ICMP is a network layer protocol. It is actually a companion to the IP protocol. The Internet Control Message Protocol version 4 (ICMPv4) has been designed to compensate for the above two deficiencies.
- ICMP messages divided into 2 types:
    1. Error-reporting message
    2. Query message

Format of ICMP messages:



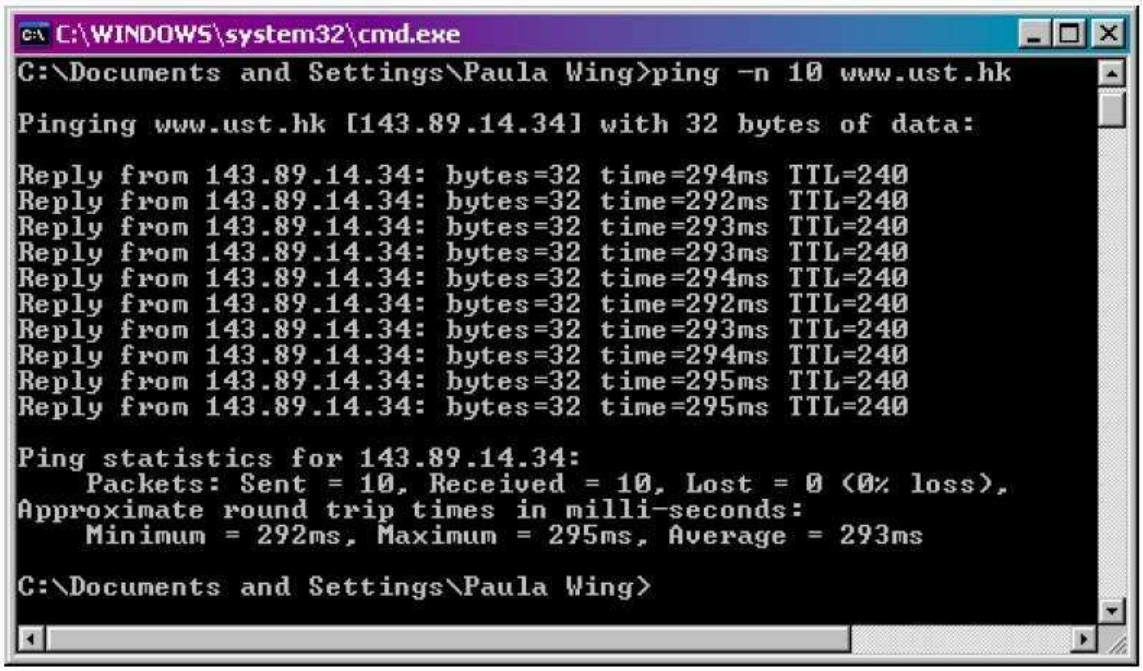
Error-reporting messages



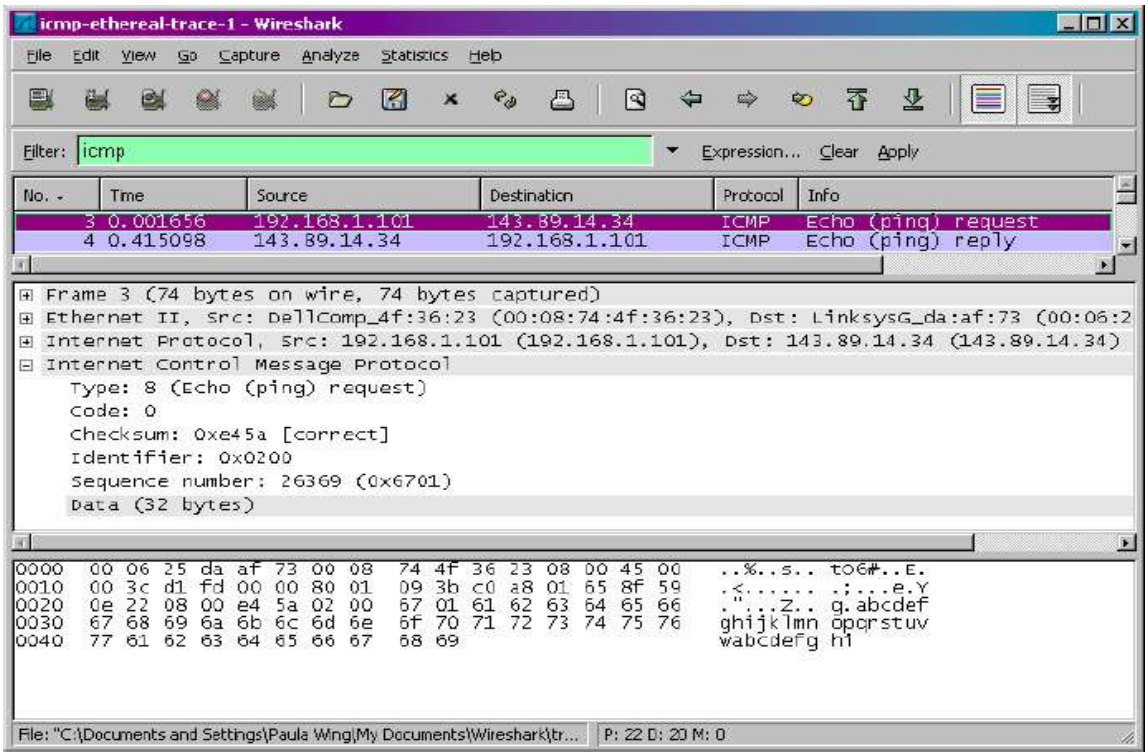
Query messages

Tracing ICMP Packets

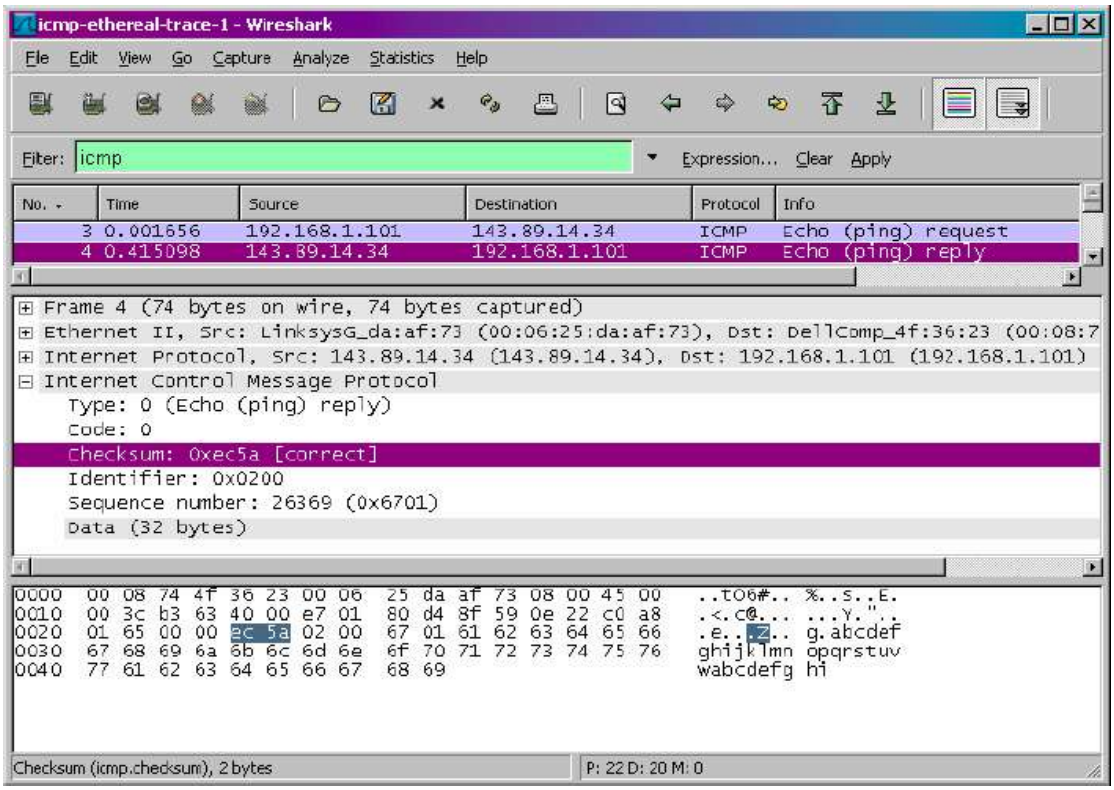
- ICMP packets are traced using Ping command here



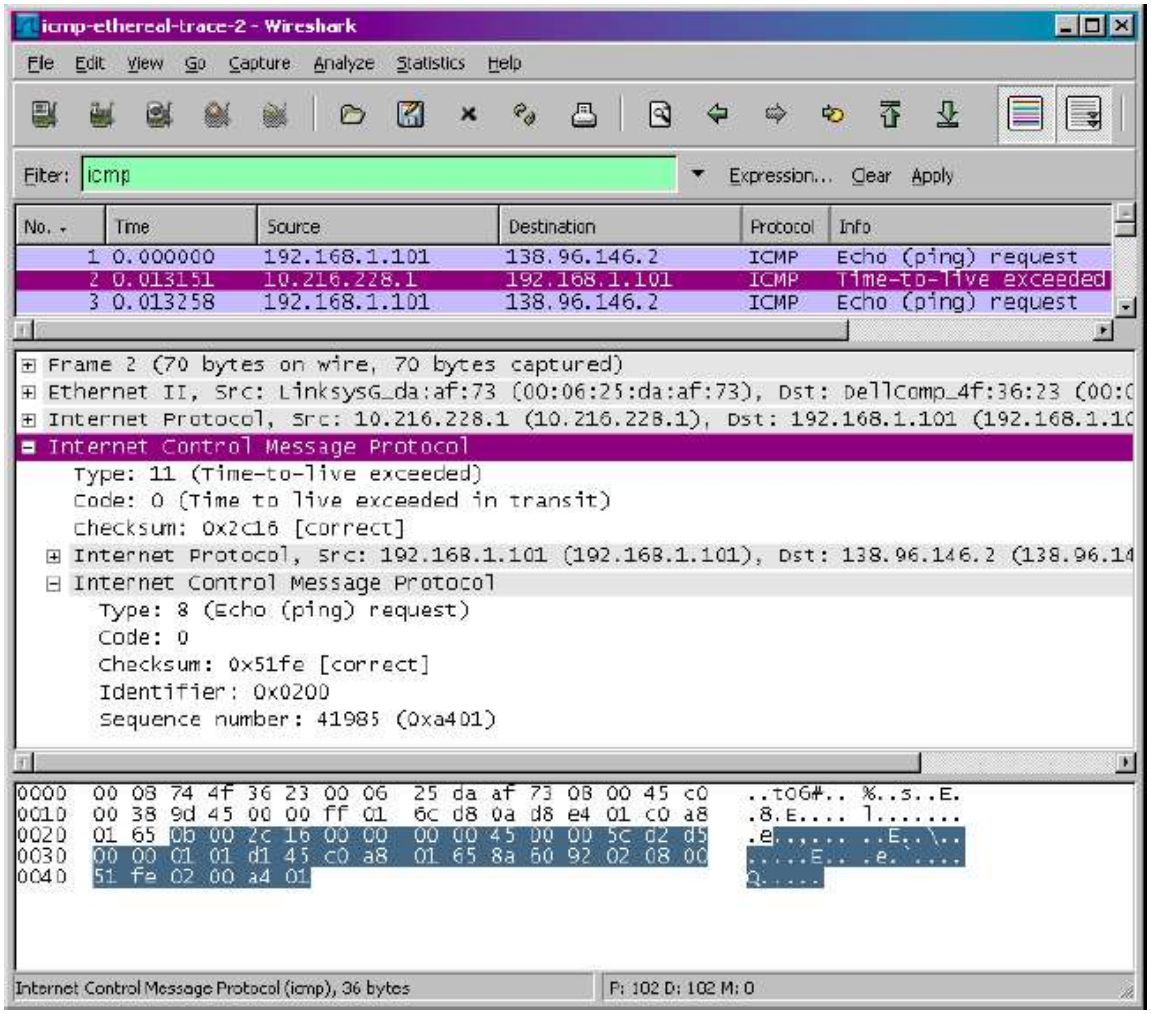
Screenshot after the PING request



ICMP Echo Request message



ICMP Echo Reply message



ICMP Error message

Result:

Thus to capture and analyzes the packet traces of DHCP and ICMP protocol using Wireshark has been completed successfully.